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Paper Code : PEC-IT601B Distributed Systems

UPID : 006590

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 x 10 = 10]

- (H) Name one primary advantage of a distributed database over a centralized one with explanation.
- (H) What are the two main types of fragmentation?
- (WH) What is the primary purpose of Concurrency Control in a DDBMS?
- (IV) What is a "Site Failure" in a distributed environment?
- (V) Define Fragmentation Transparency.
- (VI) What is the primary characteristic of a mobile database compared to a fixed one?
- (VII) What is Information in the context of data processing?
- (WHM) What is the primary purpose of Data Localization?
- (IX) Name one major cost factor considered during Distributed Query Optimization.
- (XI) What is the alternative name for the first phase of the Two-Phase Commit protocol?
- (XII) Define "Intra-query Parallelism".
- (XIII) What is the benefit of client-side caching in a mobile database environment?

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

- 2. Explain the difference between horizontal partitioning and vertical partitioning with a simple example. [5]
- 3. Explain the difference between a Shared Lock and an Exclusive Lock. [5]
- 4. Explain the basic purpose of a recovery protocol. How does it use a local log to restore consistency after a crash? [5]
- 5. Compare and contrast Shared-Disk and Shared-Nothing architectures. [5]
- 6. Compare the Top-Down design approach with the Bottom-Up design approach for distributed databases. [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

- 7. (a) Define a Parallel Database System. What are the primary goals of using such a system (e.g., performance, availability)? [5]
- (b) Explain the Shared-Memory architecture. What are its advantages and its main scalability limitations? [5]
- (c) Discuss the concept of Linear Speedup and Linear Scale-up. Why are these considered the ideal metrics for parallel systems? [5]
- 8. (a) Define a Multi-database System (MDBS). Explain the concept of "Local Autonomy" and why it is the primary challenge in MDBS design. [7]
- (b) Discuss the advantages and disadvantages of using a Multi-database approach versus a standard Distributed Database approach. [8]
- 9. (a) Describe the concept of Distributed Data Processing. How does the underlying computer network impact the performance of the database? [7]
- (b) Compare and contrast Peer-to-Peer (P2P) architecture with Client-Server architecture for distributed databases. Which is more resilient to site failure? [8]
- 10. (a) Describe the four layers of distributed query processing: Query Decomposition, Data Localization, Global Query Optimization, and Local Query Optimization. [10]
- (b) What is the objective of "Localization" in distributed data processing? Explain with a diagram. [5]
- 11. (a) [8]

Analyze the Blocking Problem in the 2PC protocol. Under what specific failure conditions does 2PC fail to reach a termination decision?

(b) Discuss the disadvantages of the 2PC protocol and explain how a Three-Phase Commit (3PC) protocol attempts to address the blocking issue.

[7]

*** END OF PAPER ***